Module 1 – Data Collection

Smart Hospital Decision Support System

1. Project Introduction

Problem Statement

Hospitals generate large volumes of patient data, but clinical information is often distributed across multiple systems. This makes it difficult for healthcare professionals to quickly assess patient risks, predict clinical outcomes, and make timely decisions regarding patient care and hospital resource utilization.

Objective

To develop a Machine Learning-based Smart Hospital Decision Support System that assists healthcare professionals by integrating multiple predictive models for:

- Patient Risk Assessment
- Hospital Readmission Prediction
- Infection Risk Prediction
- Length of Stay Prediction
- Resource Utilization Analysis

Project Modules

1. Patient Risk Assessment

2. Hospital Readmission Prediction

3. Infection Risk Prediction

4. Length of Stay Prediction

5. Resource Utilization Dashboard

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Streamlit
- SQLite
- Power BI

2. Dataset Collection

- Source (Kaggle)
- Number of datasets: **4**

3. Dataset Overview

Dataset	Rows	Columns	Target	ML Task
diabetic_data	101,766	50	readmitted	Classification
LengthOfStay	100000	28	LengthOfStay	Regression
clinical_pneumonia_dataset	1,500	12	true_label	Classification
hospital data analysis	984	10	Outcome	Classification

4. Module Mapping

Project Module	Dataset
Patient Risk Assessment	hospital data analysis
Readmission Prediction	diabetic_data
Infection Risk Prediction	clinical_pneumonia_dataset
Length of Stay Prediction	LengthOfStay